

	resistance (i.e. amplifier) of the source.
2(a)	Momentum of an object is the product of its mass and its velocity. Its unit is kg m s^{-1}
2(b)(i)	The statement is not correct. The total momentum of the system of two bodies is still conserved even if the collision is inelastic, provided there is no external resultant force acting on the system.
2(b)(ii)	This type of collision is elastic.
2(c)	Newton's first law of motion: An object remains at rest or a state of uniform motion in a straight line if there is no external resultant force acting on it. Newton's second law of motion: The rate of change of momentum of a system is proportional to the external resultant force acting on it, and the change takes place in the direction of the force. Newton's third law of motion: When a body A exerts a force on body B, body B will exert the same type of force of equal magnitude but opposite in direction on body A.
2(d)(i)	First law: Since there is a braking force acting on the car and the driver, they cannot remain in their uniform motion in a straight line, and they are brought to a stop. Second law: Since there is a braking force acting on the car and the driver, their momenta have to be changed. The rate of change of their momenta is proportional to the external resultant forces acting on them. The braking forces act opposite to their initial momenta, so their momenta decrease to zero. Third law: The road exerts a backward braking force on the car to stop the car. The car exerts an equal force on the road forward. The car exerts a backward force on the driver to stop his motion. The driver exerts an equal force on the car forward.
2(d)(ii)	Using $v^2 = u^2 + 2as$, $a = \frac{v^2 - u^2}{2s} = \frac{0^2 - 27^2}{2(60)} = -6.075 \text{ m s}^{-2}$

No.	Solution
	The braking force $F = ma = (1100)(6.075)$ $= 6682.5 = 6680 \text{ N}$ OR Work done by braking force = change in KE $-Fs = 0 - \frac{1}{2}mv^2$ $F = \frac{mv^2}{2s} = \frac{(1100)(27)^2}{2(60)} = 6680 \text{ N}$
2(d)(iii)	By Newton's second law of motion, the rate of change of momentum of the driver is proportional to the force acting on him. In a collision, the air bag serves to increase the time for the driver to reduce its momentum to zero, and hence reduces the force acting on the driver.
2(e)	As carriage A comes down, it loses gravitational potential energy. As carriage B